

Lab_Week1_monday

July 6, 2026

```
[2]: #Lab1
# Python is one of the most popular languages for large data processing
    ↳efficiently with lost of
#inbuilt functions available
print("Welcome to Data Engineering with Python")
```

Welcome to Data Engineering with Python

```
[3]: # Lab2
print("Step1: Read data")
print("Step2: Clean data")
print("Step3: Write data")
```

Step1: Read data
Step2: Clean data
Step3: Write data

```
[10]: # Lab3

customer_name = "Asha Patel"
customer_id = 101
active = True

print(f"Customer {customer_id}:{customer_name}(Active: {active})")
```

Customer 101:Asha Patel(Active: True)

```
[11]: # Lab4
A1 = 250
A2 = 99.95
A3 = "Mumbai"
A4 = False

print(f"Value: {A1}, Type: {type(A1)}")
print(f"Value: {A2}, Type: {type(A2)}")
print(f"Value: {A3}, Type: {type(A3)}")
print(f"Value: {A4}, Type: {type(A4)}")
```

Value: 250, Type: <class 'int'>
Value: 99.95, Type: <class 'float'>

Value: Mumbai, Type: <class 'str'>

Value: False, Type: <class 'bool'>

```
[13]: #Lab5
mark1 = 78
mark2 = 85.5
total = mark1+mark2
print(f"Total marks: {total}")

M1 = int("78")
M2 = float("85.5")
T = M1+M2
print(f"Total marks :{T} ")
```

Total marks: 163.5

Total marks :163.5

```
[16]: #Lab6

item_price = 249.99
quantity = 3
tax_rate = 0.05

subtotal = item_price*quantity
tax = subtotal*tax_rate
grand_total = subtotal+tax

print(f"Grand Total :{grand_total:.2f}")
```

Grand Total :787.47

```
[19]: # Lab7
order1 = 1200
order2 = 999
print("Is order1 > order2?", order1 > order2)
print("Are order1 and order2 equal?", order1 == order2)
```

Is order1 > order2? True

Are order1 and order2 equal? False

```
[24]: # Lab8
price = 1500
in_stock = True
HighV_And_Avail = (price > 1000) & in_stock
HighV_or_OOFStock = (price > 1000) | (not in_stock)

print("High value and available?", HighV_And_Avail)
print("High value or out of stock?", HighV_or_OOFStock)
```

High value and available? True
High value or out of stock? True

```
[28]: # Lab9
print("message")

name = input("Enter customer name: ")
amount = float("350.5")
print(f"Receipt -- Customer: {name}, Amount: {amount: 0.2f}")
```

message

Enter customer name: Raj

Receipt -- Customer: Raj, Amount: 350.50

```
[33]: # Lab10
product = "SSD"
price = 12999.5
qty = 2
total = price * qty
print(f"Order:{qty} x {product} @ {price:.2f} each - Total: {total:.2f}")
```

Order:2 x SSD @ 12999.50 each - Total: 25999.00

```
[41]: # Lab11

name = "Neha"
print(name) # typo

age = "30"
print("Age next year: " , int(age) + 1)

value = "123"
num = int(value)
print(num)
```

Neha

Age next year: 31

123

```
[51]: # Lab 12

m1 = "78"
m2 = "85"
m3 = "69"

total = int(m1)+int(m2)+int(m3)

percentage = total*100/300
```

```

# print(total)
# print(percentage)
print(f"Percentage : {percentage:.2f}%" )

if percentage >= 35:
    print("Result: Pass")
else:
    print("Result: Fail")

```

Percentage : 77.33%
Result: Pass

```

[54]: # Lab#13

Original_price = 4999.99
Discount_percent = 12.5

discount_amount = Original_price * (Discount_percent / 100)
final_price = Original_price - discount_amount

print(f"Discount: {discount_amount:.2f}")
print(f"Final Price: {final_price:.2f}")

```

Discount: 625.00
Final Price: 4374.99

```

[56]: # Lab14

# Blind student / customer details
name = "Rohan Verma"
id_str = "205"
item = "Textbook"
unit_price_str = "450.5" # string
quantity_str = "2"      # string

# Conversions
customer_id = int(id_str)
unit_price = float(unit_price_str)
quantity = int(quantity_str)

# Calculations
subtotal = unit_price * quantity
tax_rate = 0.05
tax = subtotal * tax_rate
grand_total = subtotal + tax

# Output block

```

```
print(f"Customer ID: {customer_id}")
print(f"Customer Name: {name}")
print(f"Item: {item}")
print(f"Quantity: {quantity}")
print(f"Subtotal: {subtotal:.2f}")
print(f"Tax (5%): {tax:.2f}")
print(f"Grand Total: {grand_total:.2f}")
```

Customer ID: 205
Customer Name: Rohan Verma
Item: Textbook
Quantity: 2
Subtotal: 901.00
Tax (5%): 45.05
Grand Total: 946.05

[]: